# Final Use-Case and Design Models

## ParkChoiceSG

Software Requirements Specification supporting material

Final implementation baseline | 8 November 2026

This document supplies the use-case descriptions, class relationships, dialog maps, architecture and request sequences for the final implementation. It is intended for inclusion in the final SRS and design documentation.

Implemented, partial/prototype and deferred requirements are identified explicitly. Planned behavior is not presented as a completed capability.

## Reading order

1. Scope and requirement inventory
2. Complete use-case descriptions (UC-01 to UC-19)
3. Use-case, class, data, architecture, sequence and UI-state diagrams
4. Scoring, evidence fields and implementation traceability

# Final scope and model conventions

ParkChoiceSG compares nearby carparks using routed driving/walking measurements, source-backed stay prices and the probability of at least one free HDB car lot at arrival. Drivers are anonymous. Operator access is a separate configured-credential check.

The working recommendation is implemented in the root backend/, parkchoice_flutter/, data-pipeline/ and ml/ directories. Experimental LTA/URA models are not enabled in application probability ranking. Older lab artifacts remain historical design material.

| Label | Meaning |
|---|---|
| Implemented | Connected application or collection behavior with the stated service, source and model limits. |
| Partial / prototype | A real control/component exists, but effects may be local, sample-based or incomplete. |
| Deferred | The requirement is retained for traceability; its intended behavior is not implemented. |

Diagram conventions

UML classes identify actual class names, values and Protocols. Module functions and database records are labelled separately. Solid links represent associations; dotted arrows represent dependencies; interface realization uses a dashed hollow triangle. Multiplicities describe actual ownership/cardinality. Key classes are shown in detail; secondary fields and helper methods may be omitted.

Dialog transitions use event [guard] / action where applicable. Sample or screen-local effects are stated on the diagram. Navigation, search and parking-session creation are separate actions.

The native source for every diagram is supplied as PlantUML, with SVG and PNG exports. Landscape diagram pages retain legible labels; use-case tables use portrait pages.

# Use-case inventory

| ID | Use case | Final status |
|---|---|---|
| UC-01 | Request carpark recommendation | Implemented with stated coverage |
| UC-02 | Inspect carpark information | Implemented in result cards |
| UC-03 | Open external navigation | Implemented |
| UC-04 | Submit error report | Deferred |
| UC-05 | Authenticate operator | Implemented credential check and app gate |
| UC-06 | Manage user error reports | Prototype interface only |
| UC-07 | Apply manual carpark override | Prototype interface only |
| UC-08 | View API health and data freshness | Prototype console display |
| UC-09 | Trigger manual source refresh | Prototype console action |
| UC-10 | Collect and log availability | Implemented separate process |
| UC-11 | Manage preferences | Partial: appearance persistence; other controls local |
| UC-12 | Manage saved locations and recent searches | Partial saved-location interface |
| UC-13 | Manage favourites | Deferred |
| UC-14 | Start parking session | Partial internal session component |
| UC-15 | Extend parking session | Prototype modal only |
| UC-16 | End parking session | Partial in-memory behavior |
| UC-17 | View session history | Prototype/sample history |
| UC-18 | Erase local data | Partial control; complete erasure deferred |
| UC-19 | View carparks on map | Implemented marker map |

UC identifiers are preserved from the earlier requirements. Partial/deferred requirements remain visible rather than being silently removed.

# UC-01 - Request carpark recommendation

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Implemented with stated coverage |
| Goal / actors | Compare nearby carparks for the intended arrival, stay and ranking preference.<br>Actors: Anonymous driver; OneMap place/routing service; HDB inference service; parking database |
| Priority / frequency | High / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The driver has selected origin and destination coordinates. A valid stay duration is set. The backend, database and required external services are reachable. |
| Main flow | 1. Open Search without creating an account. Select both places using typed search, or obtain one foreground device fix for the origin.<br>2. Choose stay duration and Best overall, Prioritise cheaper or Prioritise closer; select Find carparks.<br>3. Validate inputs. Read the cleaned location view using a destination bounding box and facility/tariff filters; calculate Haversine distance and keep the three nearest candidates within 1,500 m.<br>4. For each candidate, request a driving route from origin and a walking route to destination. Derive each arrival from a common departure instant plus its driving duration.<br>5. Quote the stay at both neighbouring whole-minute arrival starts using verified schedule versions. Retain the supported range and score its higher amount.<br>6. Request and validate HDB short-mode arrival free-lot probability. Apply route, price, freshness, identity, confidence and probability gates before scoring.<br>7. Return ranked cards, score factors and exclusion reasons. Retain a separately labelled current-reading fallback when its rules permit. |
| Alternatives / exceptions | 1. Missing inputs, invalid values or a device fix older than two minutes: request correction without submitting an invalid search.<br>2. No route, unsupported price, stale evidence or invalid forecast: exclude that candidate and retain its reason. Do not automatically fetch replacement candidates.<br>3. Missing/unsupported forecast with an allowed fallback reason: show a separate fresh-current list. A known low probability cannot bypass the forecast gate through fallback.<br>4. No usable candidates or service failure: display an unavailable/error message and allow retry. Duplicate or late UI completions do not replace a newer screen state. |
| Postconditions | A supported ordered comparison, a labelled fallback, or a reasoned unavailable state is displayed; no bay is reserved. |
| Special requirements / assumptions | Default shortlist: 3 within 1.5 km. API permits candidate_limit 1-5 and radius 100-3,000 m. Duration is 1-1,440 whole minutes. Enabled probability ranking is HDB short mode at 15-90 minutes and p_free >= 0.90. Saved-place integration, vehicle/walking preference filtering and effective operator overrides are deferred. |

# UC-02 - Inspect carpark information

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Implemented in result cards |
| Goal / actors | Understand the evidence and trade-offs of a returned carpark.<br>Actors: Anonymous driver |
| Priority / frequency | High / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | A recommendation or current-reading fallback response has been received. |
| Main flow | 1. Inspect the returned carpark card and canonical identity/name.<br>2. Read the full-stay cost or supported cost range, driving duration, walking distance, current reading and forecast where available.<br>3. Read the recommendation basis and unavailable/exclusion explanations. Open the map to inspect locations if required. |
| Alternatives / exceptions | 1. A field lacks supported evidence: display its unavailable status instead of a fabricated zero or forecast.<br>2. An outdated or malformed response is rejected by the client response checks; retain a recoverable error state. |
| Postconditions | The driver can compare the displayed evidence without modifying provider or database records. |
| Special requirements / assumptions | The current implementation uses result cards rather than a separate complete CarparkDetailUI. A future probability is not an expected free-lot count. |

# UC-03 - Open external navigation

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Implemented |
| Goal / actors | Hand off the selected carpark coordinates for driving navigation.<br>Actors: Anonymous driver; device operating system; external navigation application/browser |
| Priority / frequency | High / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | A valid carpark marker is selected in the map view. |
| Main flow | 1. Select the carpark marker and choose external navigation.<br>2. Validate WGS84 destination coordinates and construct a Google Maps directions URL.<br>3. Ask the operating system to open the URL externally; the external application manages its own origin, location permission and navigation. |
| Alternatives / exceptions | 1. Missing or invalid coordinates: keep the map accessible and show a failure message.<br>2. The operating system rejects the handoff or no handler opens: display an error and permit retry.<br>3. Return from navigation: resume the existing app screen. |
| Postconditions | The URL opens externally or the driver receives an error; no parking session or reservation is created. |
| Special requirements / assumptions | OneMap supplies in-app map tiles and route measurements. External Google Maps URL navigation uses no embedded Google SDK or paid Google routing API. Coordinates are not verified entrances. |

# UC-04 - Submit error report

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Deferred |
| Goal / actors | Submit a source-data issue linked to a carpark.<br>Actors: Anonymous driver; operator |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Intended: a carpark is identified and reporting is available. |
| Main flow | 1. Intended: open a report form from the selected carpark.<br>2. Enter issue category/details; validate the carpark and input.<br>3. Persist the report and display its real reference number. |
| Alternatives / exceptions | 1. Intended: invalid input or unknown identity retains the form and requests correction.<br>2. Intended: save failure or quota rejection displays the actual error without claiming submission. |
| Postconditions | Intended: a persisted report is available for operator review. |
| Special requirements / assumptions | No driver report-submission form or persistence endpoint is connected in the final baseline. This description records the deferred requirement and is excluded from implemented use-case views. |

# UC-05 - Authenticate operator

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Implemented credential check and app gate |
| Goal / actors | Open the operator console after credential verification.<br>Actors: Operator; recommendation API |
| Priority / frequency | High / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The operator opens Admin. Server-side OPERATOR_USERNAME and OPERATOR_PASSWORD_HASH are configured. |
| Main flow | 1. Display OperatorLoginScreen unless this app process already has a signed-in operator state.<br>2. Validate nonempty fields and submit POST /operator/login with username and password.<br>3. Verify the credential against the configured PBKDF2-SHA256 hash using constant-time comparisons.<br>4. On HTTP 200, if the login screen is still active, set OperatorSession.signedIn and open AdminPanelScreen. |
| Alternatives / exceptions | 1. Empty input: reject locally. Incorrect credentials return 401; schema-invalid input returns 422.<br>2. Missing/malformed server configuration returns 503. Timeout/network failure keeps the operator on login with an error.<br>3. Repeated taps send one request; leaving the login screen discards a late success.<br>4. Sign out clears the in-memory flag; app restart also clears it. |
| Postconditions | The app console gate is open for this process or remains closed. |
| Special requirements / assumptions | This credential endpoint does not issue a server token/session or authorize administrative writes. No OperatorAccount database table or driver account system is implemented. |

# UC-06 - Manage user error reports

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Prototype interface only |
| Goal / actors | Inspect and resolve displayed sample alerts.<br>Actors: Operator |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The app operator gate is open. |
| Main flow | 1. Open the sample alert list in the console.<br>2. Inspect an alert and choose its resolve action.<br>3. Update the alert status in the current screen state and display feedback. |
| Alternatives / exceptions | 1. Leaving/recreating the screen may restore its sample initial state. |
| Postconditions | Only the current console state changes. |
| Special requirements / assumptions | Alerts are fixed sample records. A persisted report queue, reviewer audit and server-side resolution are deferred. |

# UC-07 - Apply manual carpark override

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Prototype interface only |
| Goal / actors | Demonstrate local override controls.<br>Actors: Operator |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The app operator gate is open. |
| Main flow | 1. Open Add override and choose the displayed carpark/status and optional reason.<br>2. Confirm to add a local override model to the screen list.<br>3. Remove a displayed override using its local action if required. |
| Alternatives / exceptions | 1. Cancel or close the modal: do not add a record.<br>2. Repeated close/submit actions and completion after leaving the screen are ignored. |
| Postconditions | The current console list changes without changing recommendation inputs. |
| Special requirements / assumptions | No override database write, tariff replacement, effective-date reconciliation or closure filter is connected. Closure and rate correction remain separate deferred business rules; neither is applied by the sample console. |

# UC-08 - View API health and data freshness

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Prototype console display |
| Goal / actors | Inspect the console health/freshness layout.<br>Actors: Operator |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The app operator gate is open. |
| Main flow | 1. Open the console data-source section.<br>2. Inspect its displayed health, request, failure and cache-age fields. |
| Alternatives / exceptions | 1. The displayed records remain sample values rather than results of a feed-health query. |
| Postconditions | No provider or database record changes. |
| Special requirements / assumptions | The console is not wired to live monitoring. GET /health reports backend process liveness only; it is not proof that providers, database, tariffs and model are all healthy. |

# UC-09 - Trigger manual source refresh

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Prototype console action |
| Goal / actors | Demonstrate the manual-refresh control.<br>Actors: Operator |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The app operator gate is open and a sample data source is displayed. |
| Main flow | 1. Select the source refresh control.<br>2. Display local action feedback. |
| Alternatives / exceptions | 1. Navigation away ends the interaction without starting collection. |
| Postconditions | No provider polling or database ingestion is started by this control. |
| Special requirements / assumptions | A protected refresh endpoint and auditable refresh job are deferred. Actual refresh is performed by the separate collection scripts under UC-10. |

# UC-10 - Collect and log availability

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Implemented separate process |
| Goal / actors | Maintain timestamped, cleaned parking observations and source evidence.<br>Actors: Collection operator or scheduler; HDB, LTA and URA feeds; parking database |
| Priority / frequency | High / Configured interval; default every five minutes |
| Trigger | The collection operator/scheduler starts a configured capture. |
| Preconditions | The collection process has private provider/database configuration and network access. |
| Main flow | 1. Start collect_live.py for one cycle or collect_periodic.py for configured repeated availability cycles.<br>2. Fetch provider responses and archive compressed raw captures with run IDs, timestamps and hashes.<br>3. Validate identifiers/counts/timestamps and preserve rejected rows or quality flags. Map exact agency/source identities.<br>4. Load accepted reference, metadata and availability observations in transactions without replacing historical observations.<br>5. For the price-capable live collector, capture and compile URA rate details when due, preserving unsupported windows.<br>6. For periodic collection, record cycle results and continue at the configured start-to-start interval; skip elapsed slots rather than creating catch-up bursts. |
| Alternatives / exceptions | 1. Missing credentials or provider failure: stop/report the affected cycle using its manifest/log rather than load a fabricated success.<br>2. Duplicate periodic process in the same workspace: reject it using the collector lock.<br>3. Invalid counts, missing references or stale source observations retain explicit quality outcomes. |
| Postconditions | Accepted data and provenance are stored; historical evidence remains available for exports and comparison. |
| Special requirements / assumptions | Periodic default: 5 minutes. HDB observation times differ from LTA/URA retrieval proxies. The model service independently maintains recent HDB history; recommendation requests do not start collection or training. Periodic availability collection does not approve HDB/commercial tariffs. |

# UC-11 - Manage preferences

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Partial: appearance persistence; other controls local |
| Goal / actors | Choose appearance and inspect preference controls.<br>Actors: Anonymous driver; device preference storage |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | The app is running; no driver account is required. |
| Main flow | 1. Open Preferences and select Light, Dark or System appearance.<br>2. Apply the theme immediately and save appearance_mode using SharedPreferences.<br>3. Optionally change vehicle and walking controls in the current screen. Select ranking mode on Search for actual recommendation scoring. |
| Alternatives / exceptions | 1. Reload the app: restore saved appearance.<br>2. Leave/recreate Preferences: other controls may return to their initial local values. |
| Postconditions | Appearance is persisted locally; other controls are local screen state. |
| Special requirements / assumptions | Vehicle/height/walking-tolerance controls are not sent to the recommendation API. The three search modes are implemented but are not a persisted preference profile. |

# UC-12 - Manage saved locations and recent searches

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Partial saved-location interface |
| Goal / actors | Maintain saved-place labels in the current Preferences screen.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Preferences is open. |
| Main flow | 1. Inspect the initial sample Home/Work entries.<br>2. Open Add location, enter a nonempty label and address, and confirm.<br>3. Add the local entry or remove an existing one from the displayed list. |
| Alternatives / exceptions | 1. Cancel or invalid/blank input: do not add the entry.<br>2. Screen recreation can restore the initial sample list. |
| Postconditions | The in-memory screen list changes. |
| Special requirements / assumptions | No durable saved-location repository, geocoded saved-place selection in Search, or recent-search history is connected. The earlier maximum-ten recent-search requirement remains deferred. |

# UC-13 - Manage favourites

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Deferred |
| Goal / actors | Keep preferred carparks for later access.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Intended: a valid carpark is displayed. |
| Main flow | 1. Intended: mark/unmark a carpark as a favourite.<br>2. Intended: store its canonical identity locally and reopen it from a favourites list. |
| Alternatives / exceptions | 1. Intended: a removed or unsupported carpark remains labelled unavailable rather than displaying stale invented measurements. |
| Postconditions | Intended: the device-local favourite list is updated. |
| Special requirements / assumptions | No connected favourites list or durable Favourite repository is implemented. |

# UC-14 - Start parking session

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Partial internal session component |
| Goal / actors | Track one active stay in memory.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | An internal caller supplies carpark name, location and hourly rate; no session is active. |
| Main flow | 1. Call ParkingSessionManager.startSession with the supplied details.<br>2. Create ParkingSession with the current start time and notify listeners.<br>3. Tracker shows elapsed time and linear running cost when an active session exists. |
| Alternatives / exceptions | 1. If a session already exists, startSession returns false without replacing it. |
| Postconditions | At most one active session is stored in process memory. |
| Special requirements / assumptions | Search/map/navigation do not automatically start a session. No complete user-facing start flow or durable session storage is connected. Linear hourly-rate display does not reuse the backend tariff engine. |

# UC-15 - Extend parking session

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Prototype modal only |
| Goal / actors | Select a proposed extension interval.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Tracker displays an active session. |
| Main flow | 1. Open Extend session and choose an interval.<br>2. Confirm or cancel the modal. |
| Alternatives / exceptions | 1. Cancel/close leaves the screen unchanged. |
| Postconditions | Confirmation displays feedback; the underlying session timing is not extended. |
| Special requirements / assumptions | No expiry adjustment, reminder scheduling, notification permission flow or backend repricing occurs. These remain deferred. |

# UC-16 - End parking session

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Partial in-memory behavior |
| Goal / actors | Clear the current tracked stay.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | An active in-memory session exists. |
| Main flow | 1. Select End session and confirm.<br>2. Create a recent-session display entry in the current tracker screen using its elapsed-time/linear-rate values.<br>3. Clear ParkingSessionManager and update the screen. |
| Alternatives / exceptions | 1. Cancel confirmation retains the active session. |
| Postconditions | The in-memory active session is cleared. |
| Special requirements / assumptions | No payment is collected. The calculated display value is not an authoritative tariff charge, and the recent entry is not durable. |

# UC-17 - View session history

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Prototype/sample history |
| Goal / actors | Inspect the tracker history layout and locally ended entries.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Tracker is open. |
| Main flow | 1. Inspect the sample grouped recent sessions.<br>2. If a session was ended in this screen instance, inspect the inserted local entry. |
| Alternatives / exceptions | 1. Screen recreation or app restart may restore sample state and lose local entries. |
| Postconditions | No persisted history is read or written. |
| Special requirements / assumptions | Durable history, historical tariff receipts and reliable aggregate expenditure are deferred. |

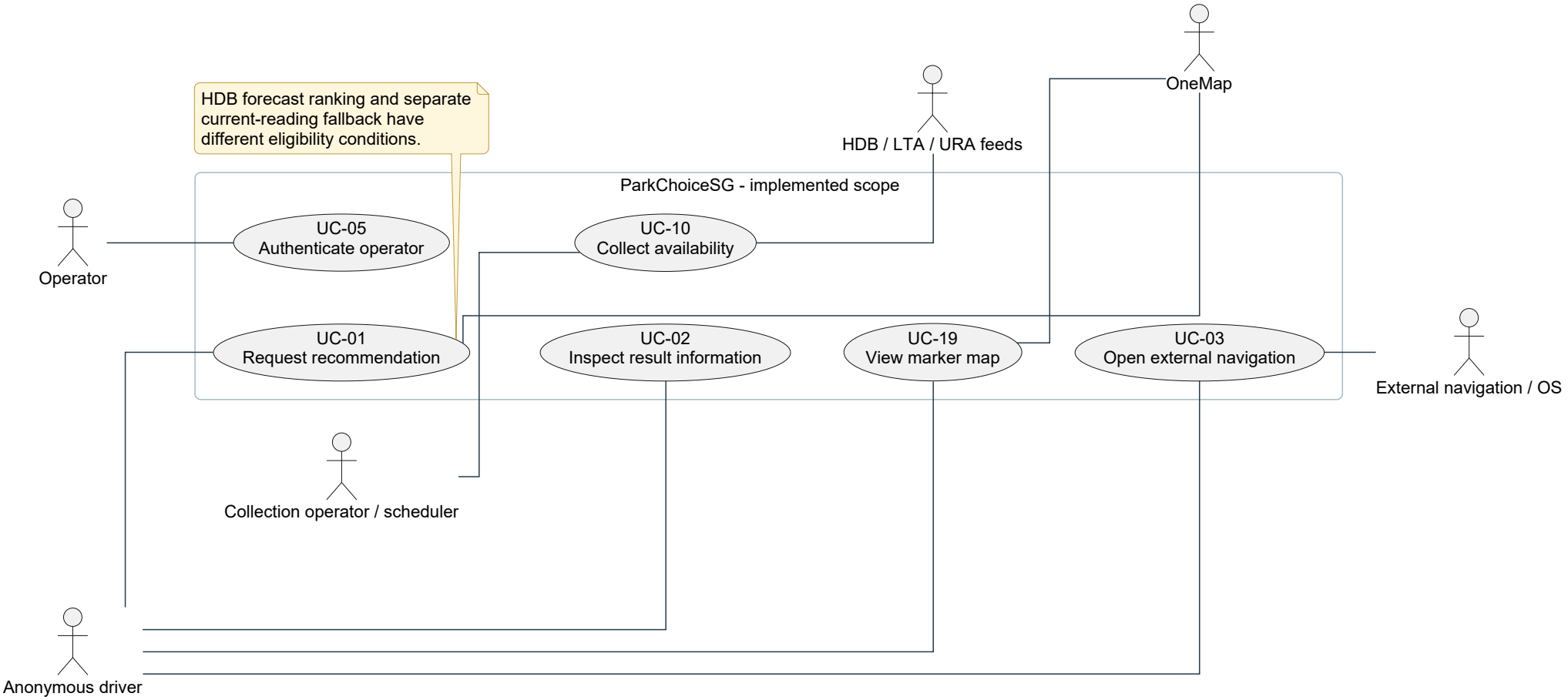
# UC-18 - Erase local data

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Partial control; complete erasure deferred |
| Goal / actors | Reset the implemented local preference controls.<br>Actors: Anonymous driver |
| Priority / frequency | Medium / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Preferences is open. |
| Main flow | 1. Open the erase confirmation.<br>2. Confirm to reset the current walking-distance and vehicle controls to their defaults. |
| Alternatives / exceptions | 1. Cancel retains existing screen state. |
| Postconditions | Those two screen controls reset. |
| Special requirements / assumptions | The current action does not erase saved appearance, saved-place sample lists, sessions or search/history repositories. It must not be treated as a complete privacy-data deletion feature. |

# UC-19 - View carparks on map

| | |
|---|---|
| Revision / owner | 8 November 2026; ParkChoiceSG |
| Status | Implemented marker map |
| Goal / actors | Inspect selected locations and returned carparks spatially.<br>Actors: Anonymous driver; OneMap tile service; device operating system |
| Priority / frequency | High / On demand |
| Trigger | The actor invokes the named action or opens its app control. |
| Preconditions | Origin, destination and result candidate coordinates are available. |
| Main flow | 1. Choose View map from Search/results.<br>2. Validate and snapshot coordinates and candidate data for the map screen.<br>3. Render attributed OneMap tiles and markers for origin, destination and returned carparks.<br>4. Select a carpark marker to inspect it or open external navigation; return to the search screen when finished. |
| Alternatives / exceptions | 1. Invalid coordinates or no usable map data: show a recoverable error.<br>2. Tile/network or navigation failure does not change the recommendation records.<br>3. Repeated navigation taps do not open duplicate map screens. |
| Postconditions | The map is displayed or a clear unavailable state is shown. |
| Special requirements / assumptions | The screen is a marker map, not embedded turn-by-turn navigation or continuous background GPS tracking. |

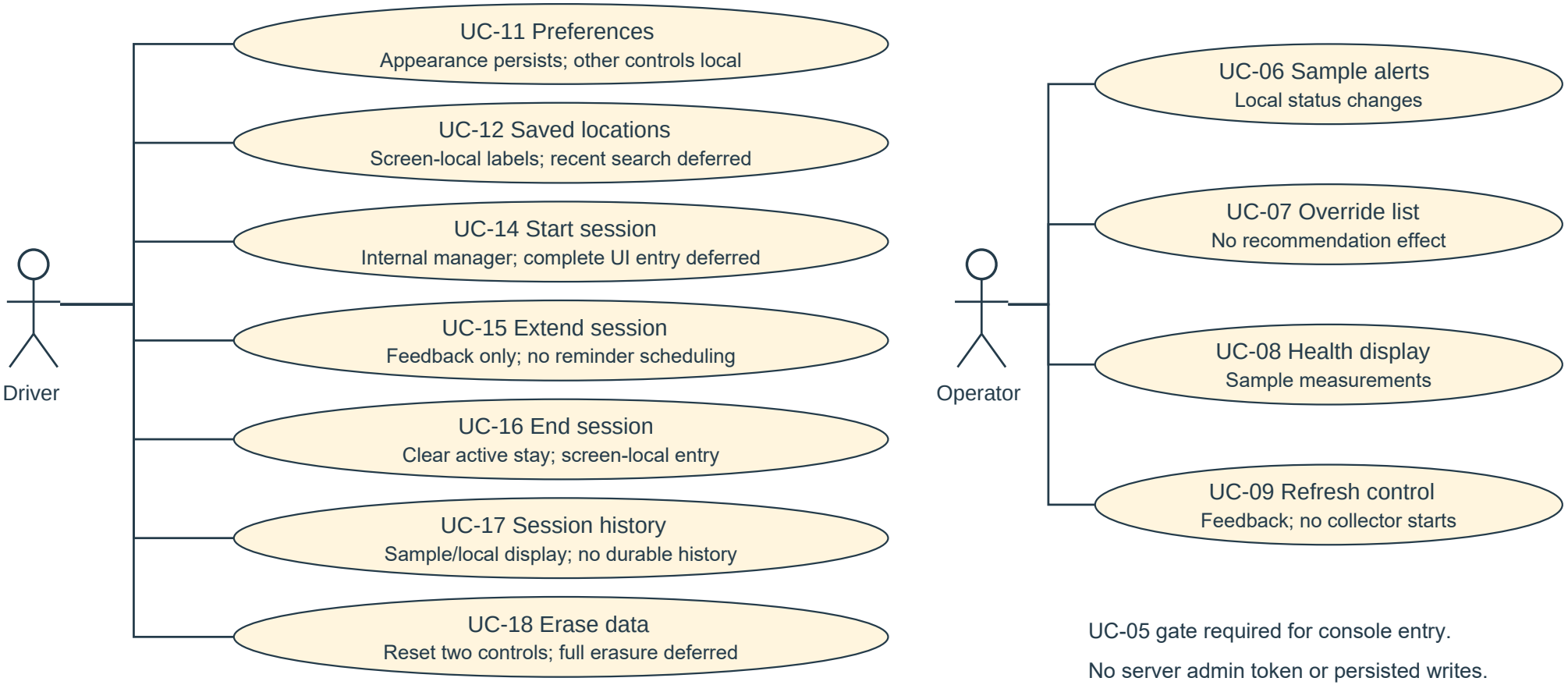
# Figure 1. Use-case model: connected application and collection




Associations show external participation. Result inspection/map/navigation require returned candidate data; they do not imply separate backend services. OneMap supplies search, routes and tiles.

Implementation reference: backend/app.py; recommendation.py; data-pipeline/collect_live.py; Flutter screens

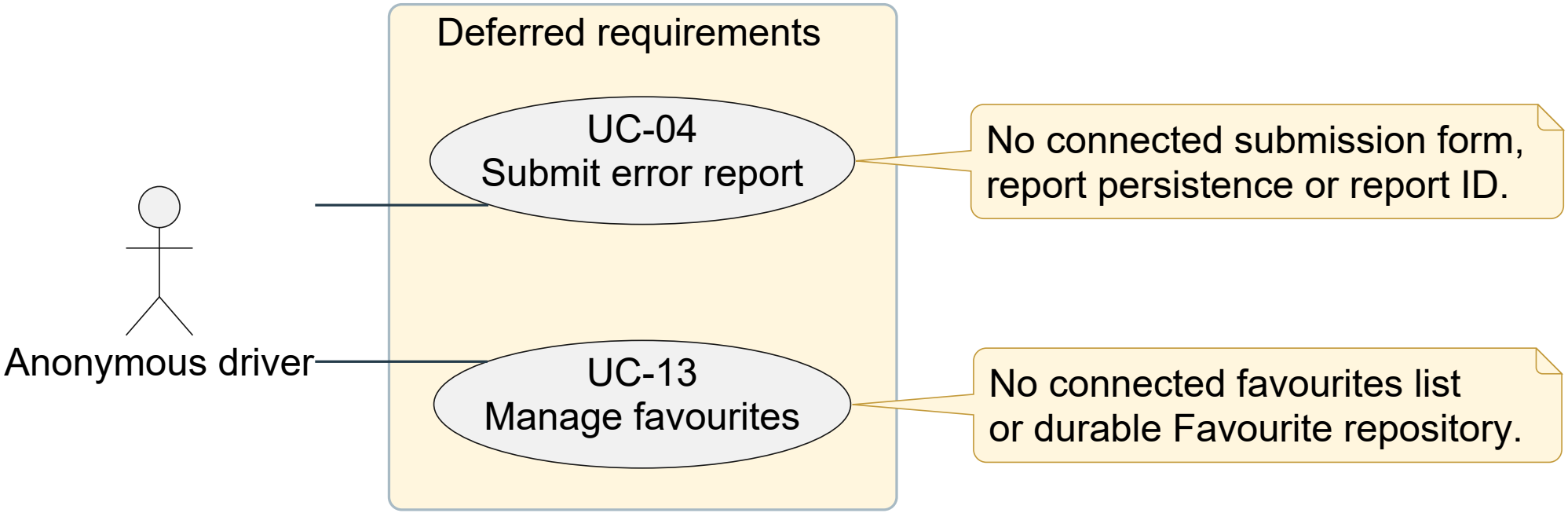
# Figure 2. Use-case model: partial and prototype functions




These functions remain in the requirements inventory with their precise implementation limits. A local control or sample record does not establish durable persistence or a provider operation.

Implementation reference: Flutter PreferencesScreen, ParkingSessionTrackerScreen, AdminPanelScreen

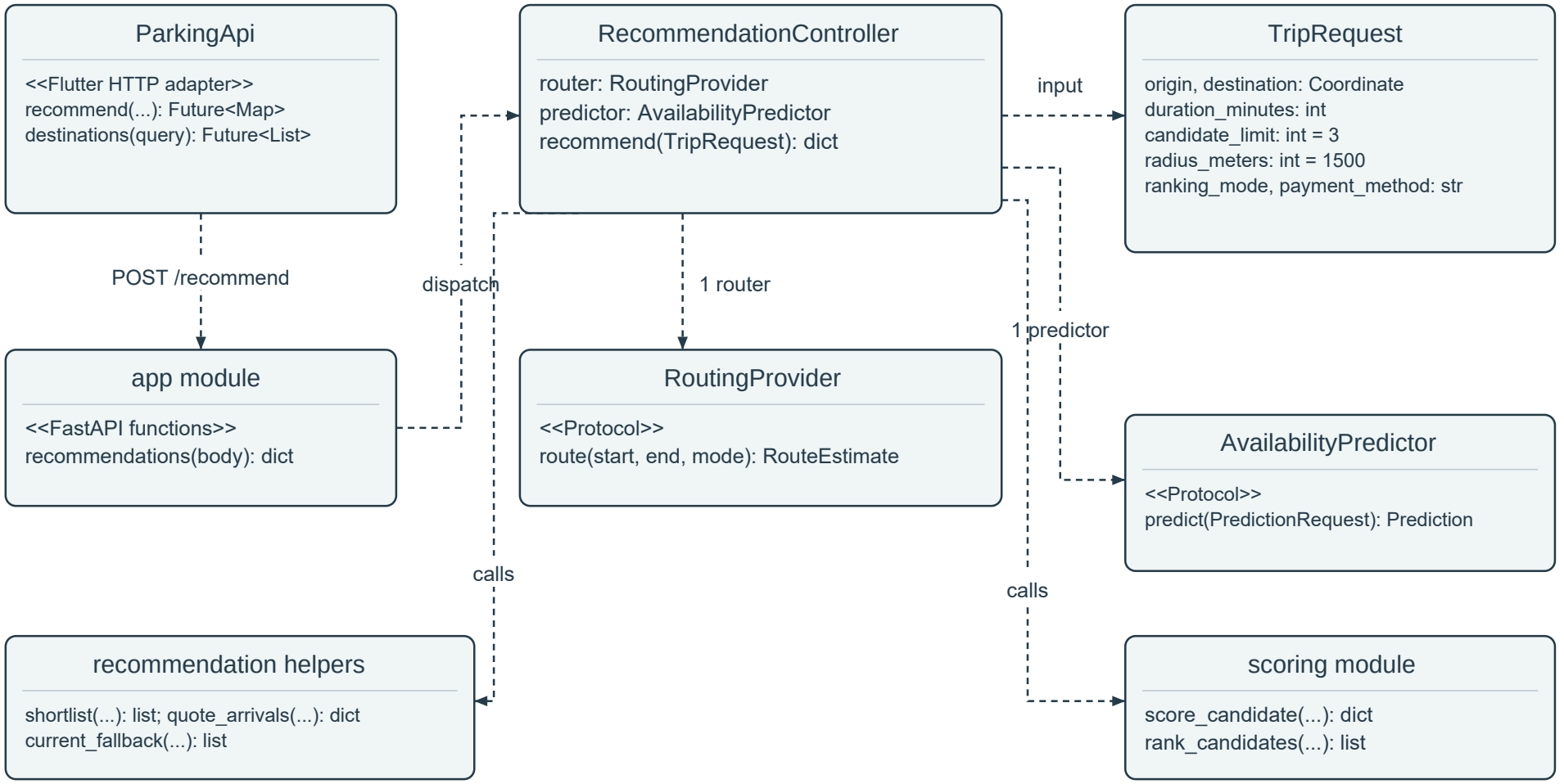
# Figure 3. Use-case model: deferred requirements




Other unfinished extensions are identified under their existing partial use cases: durable saved places/history, authenticated operator writes, complete erasure and reminder scheduling. No requirement is silently treated as completed.

Implementation reference: Requirements inventory and implementation comparison

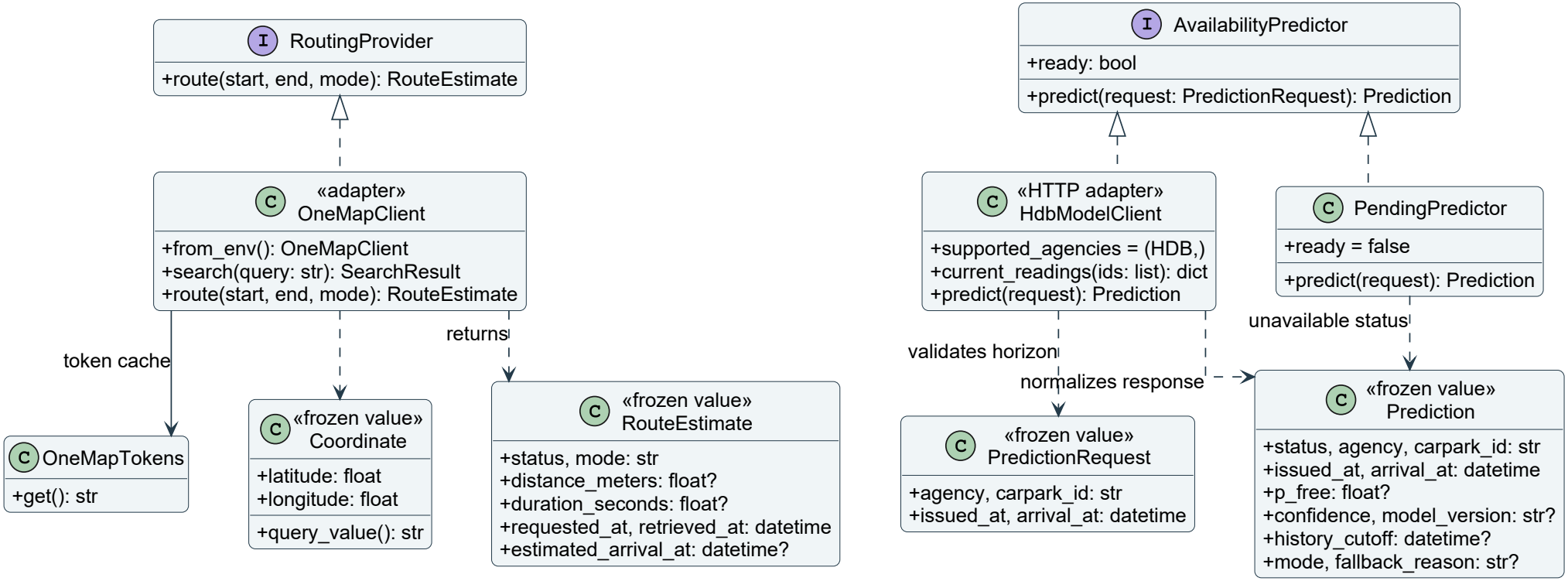
# Figure 4. Key classes: recommendation coordination




Python modules are labelled as functions rather than invented service classes. The controller receives one router and one predictor. TripRequest holds two validated Coordinate values.

Implementation reference: backend/recommendation.py; app.py; scoring.py; parking_api.dart

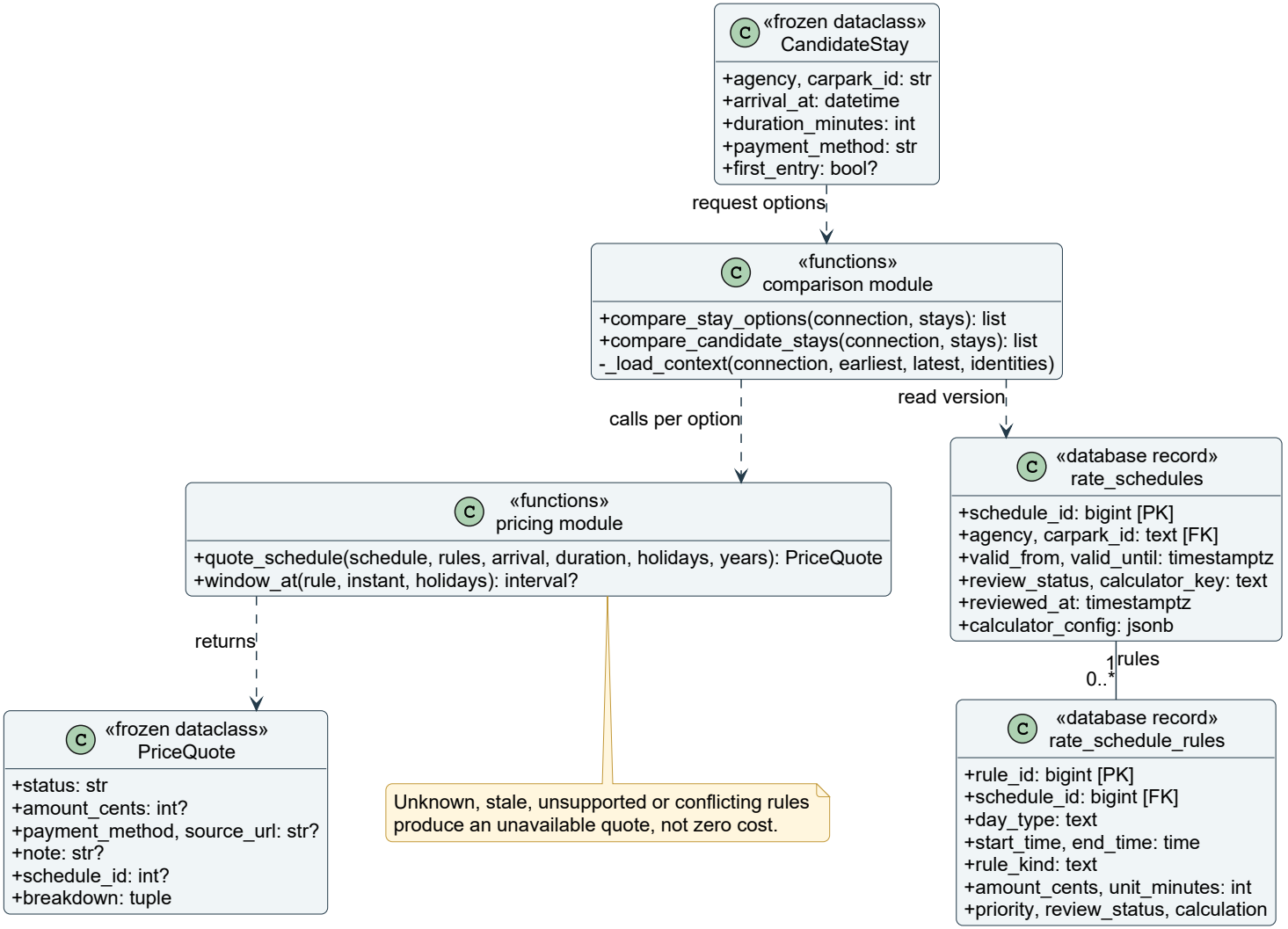
# Figure 5. Key classes: routing and prediction contracts




OneMapClient implements the routing Protocol structurally. HdbModelClient accepts only HDB short-mode probability forecasts at 15-90 minute horizons. Service failures and unsupported forecasts become explicit statuses.

Implementation reference: backend/onemap_adapter.py; onemap_auth.py; model_adapter.py; prediction.py; ml/service/

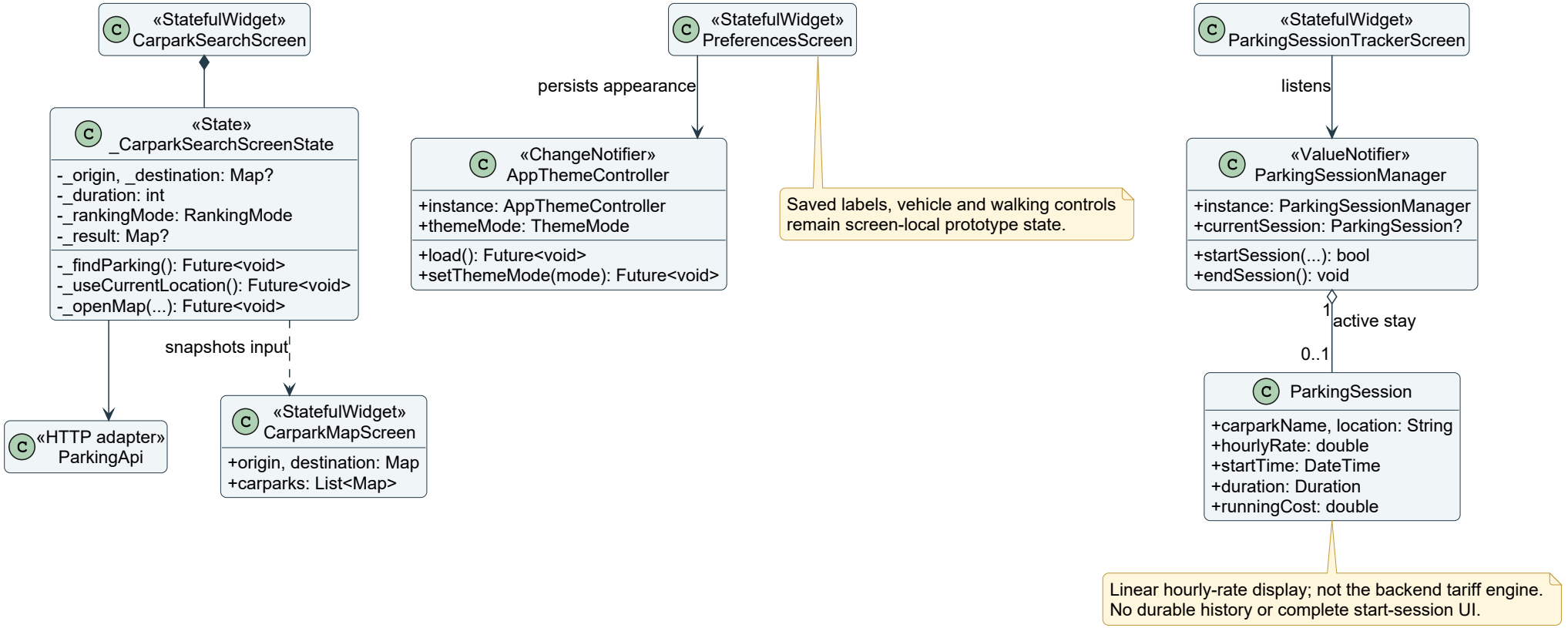
# Figure 6. Key classes and functions: stay pricing




Money is stored in integer SGD cents. The calculator applies day/time windows, priority, billing increments, grace, tiers and supported caps. Neighbouring-minute arrival quotes are compared before a cost is eligible for ranking.

Implementation reference: backend/comparison.py; pricing.py; database/rate schedule definitions

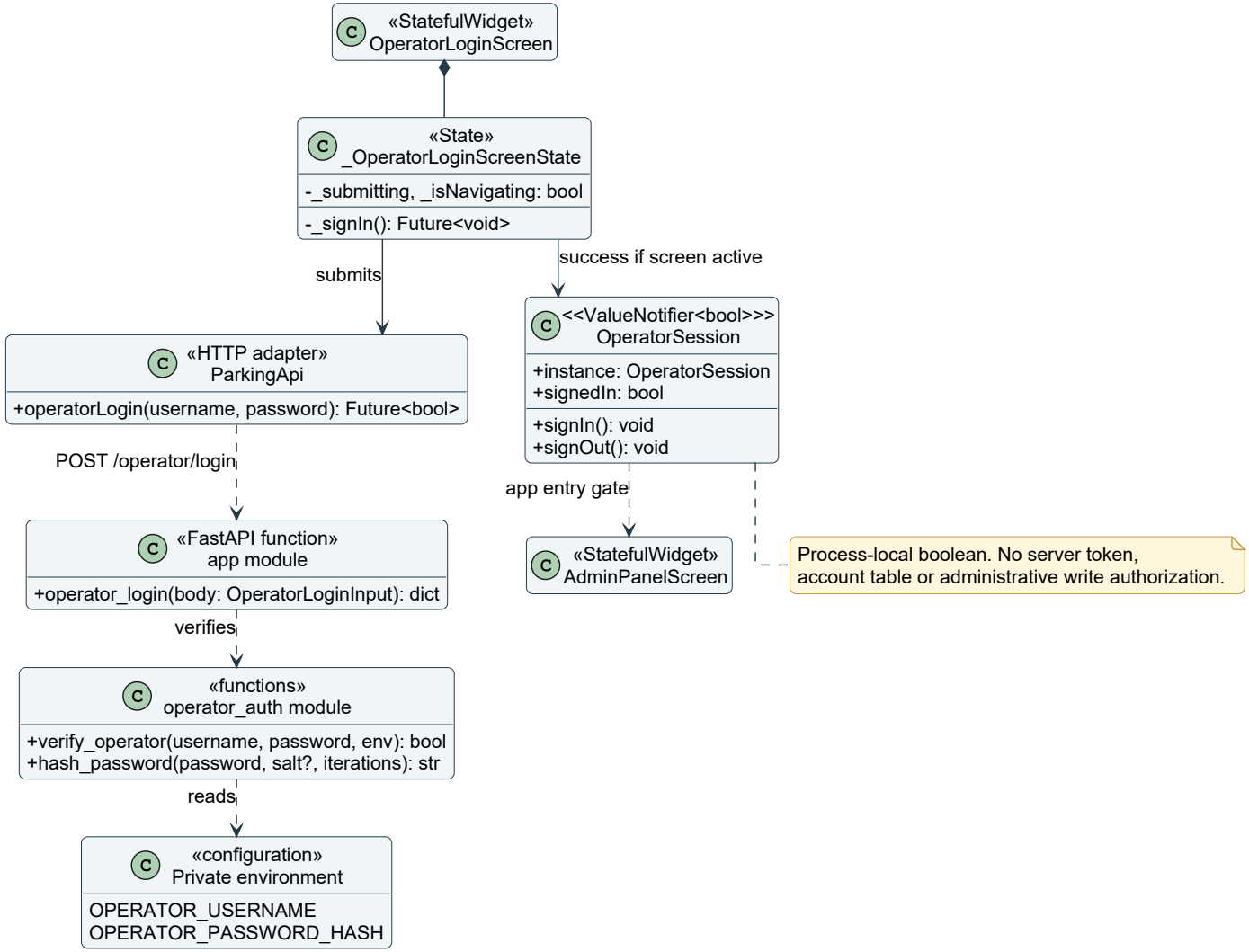
# Figure 7. Key classes: Flutter boundaries and local state




Widget classes own their State objects. The session manager holds at most one in-memory stay. Only appearance uses persistent preference storage; other local-data requirements remain partial or deferred.

Implementation reference: parkchoice_flutter/lib/screens/; services/; models/ranking_mode.dart

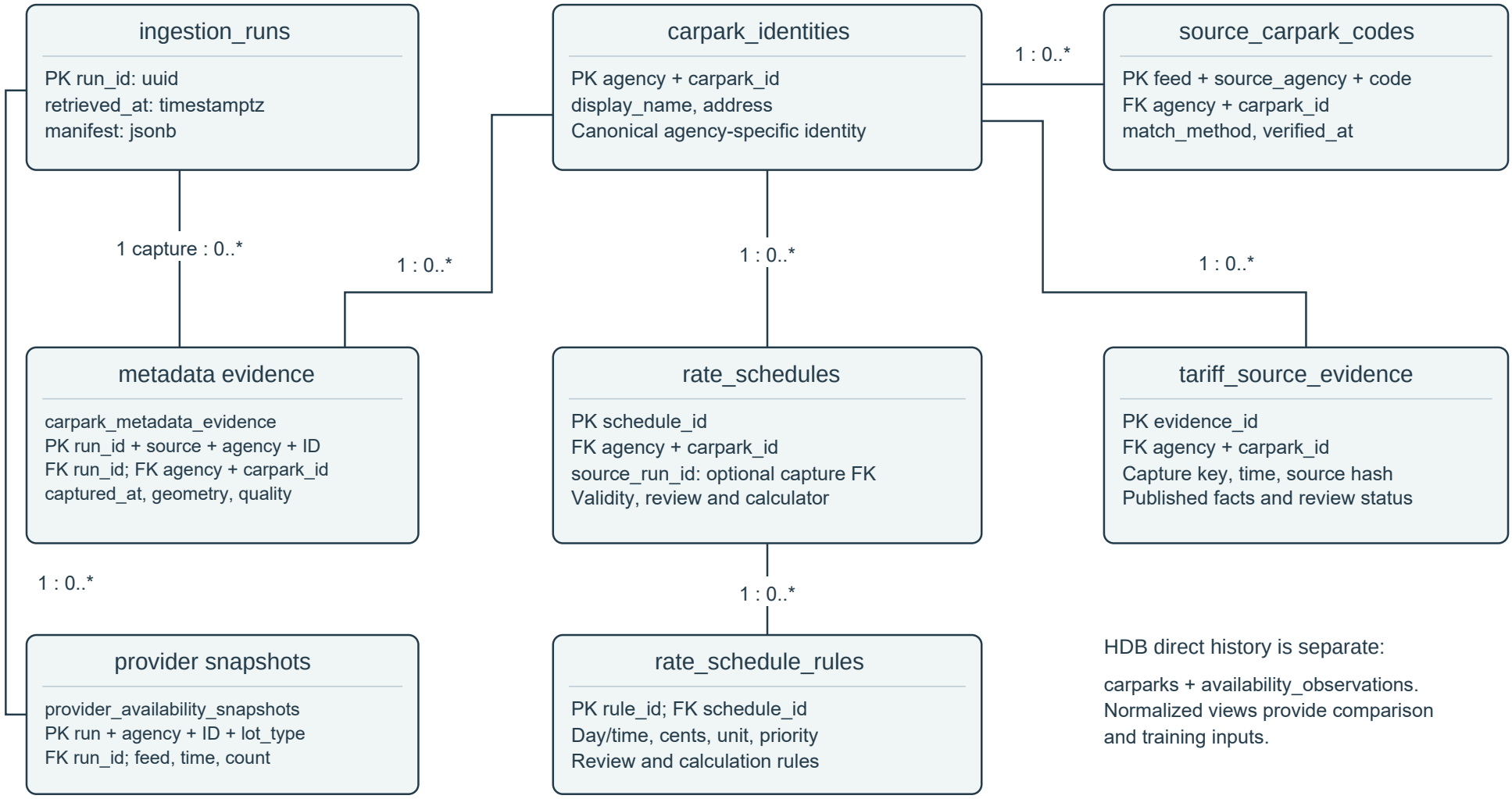
# Figure 8. Key classes: operator access




The configured credential is verified server-side. The successful response enables a client-side console entry gate for the app process. Console sample actions do not imply authenticated backend administration.

Implementation reference: operator_login_screen.dart; operator_session.dart; backend/app.py; operator_auth.py

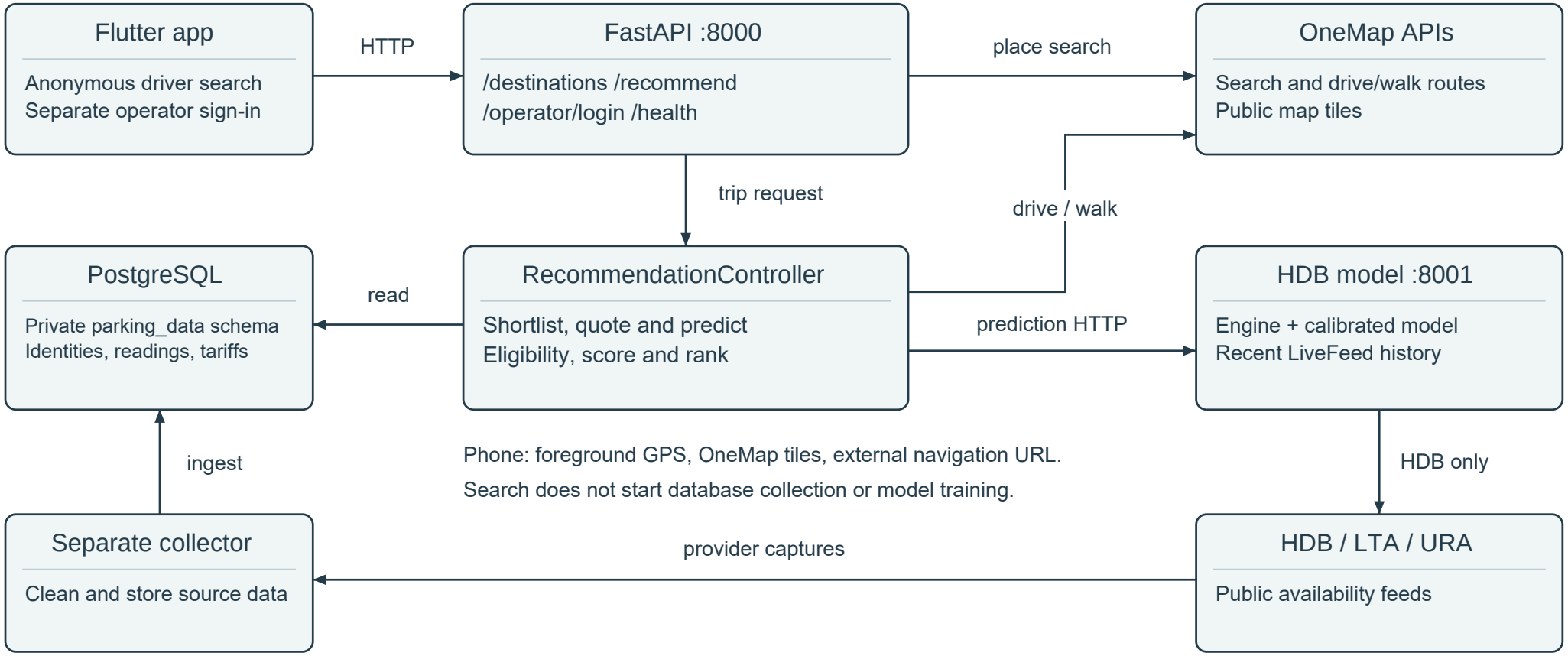
# Figure 9. Persistent data model: identity, evidence and tariffs




This selected physical-data view omits secondary columns and lookup tables. HDB direct history and provider captures remain separate provenance paths. Device preferences, session samples and operator app state are not backend driver-account tables.

Implementation reference: backend/database/schema.sql and schema_v2.sql through schema_v8.sql

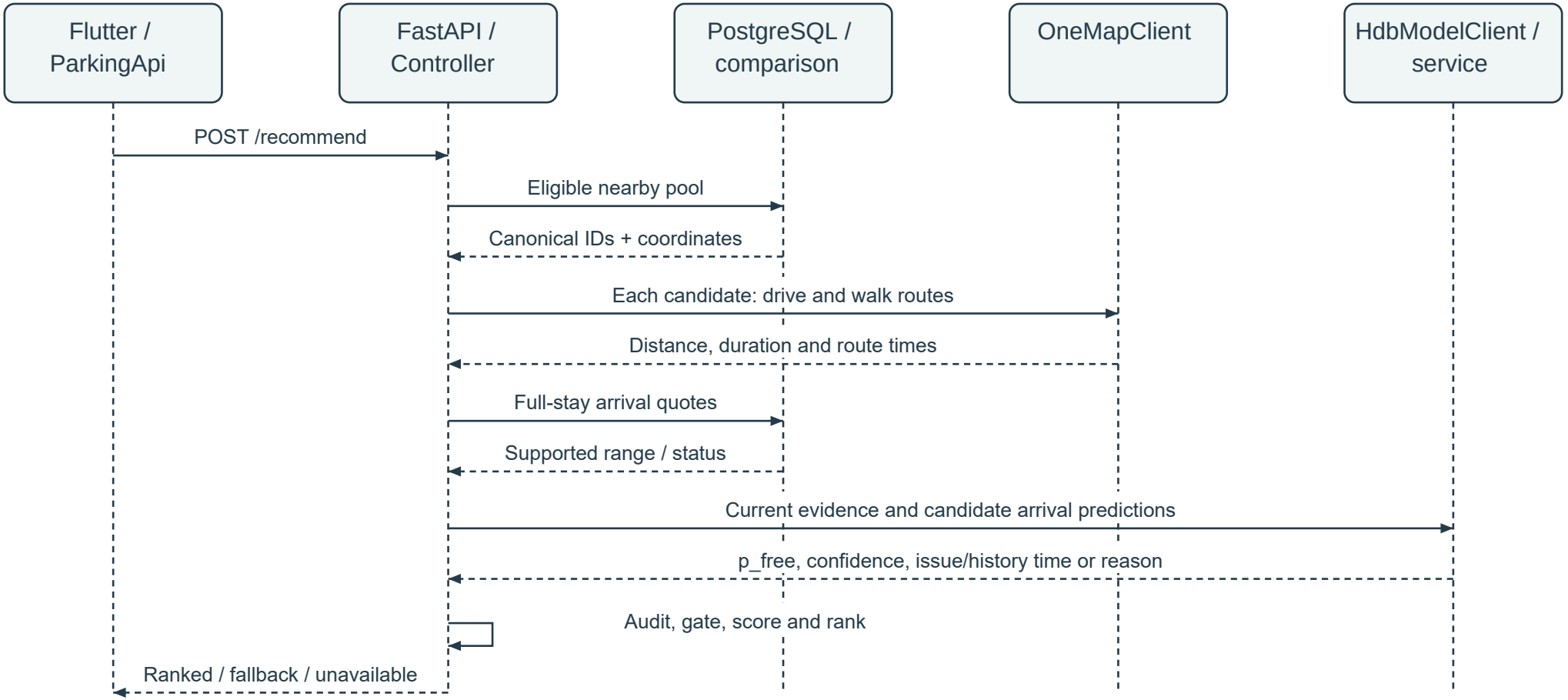
# Figure 10. System architecture and process boundaries




The inference service polls HDB into its own recent-history cache; the separate database collector maintains stored evidence and exports. Local development uses HTTP and USB port forwarding. Public deployment/authenticated admin operations are outside this baseline.

Implementation reference: backend/app.py; recommendation.py; ml/service/; data-pipeline/; Flutter app

# Figure 11. UC-01 request sequence

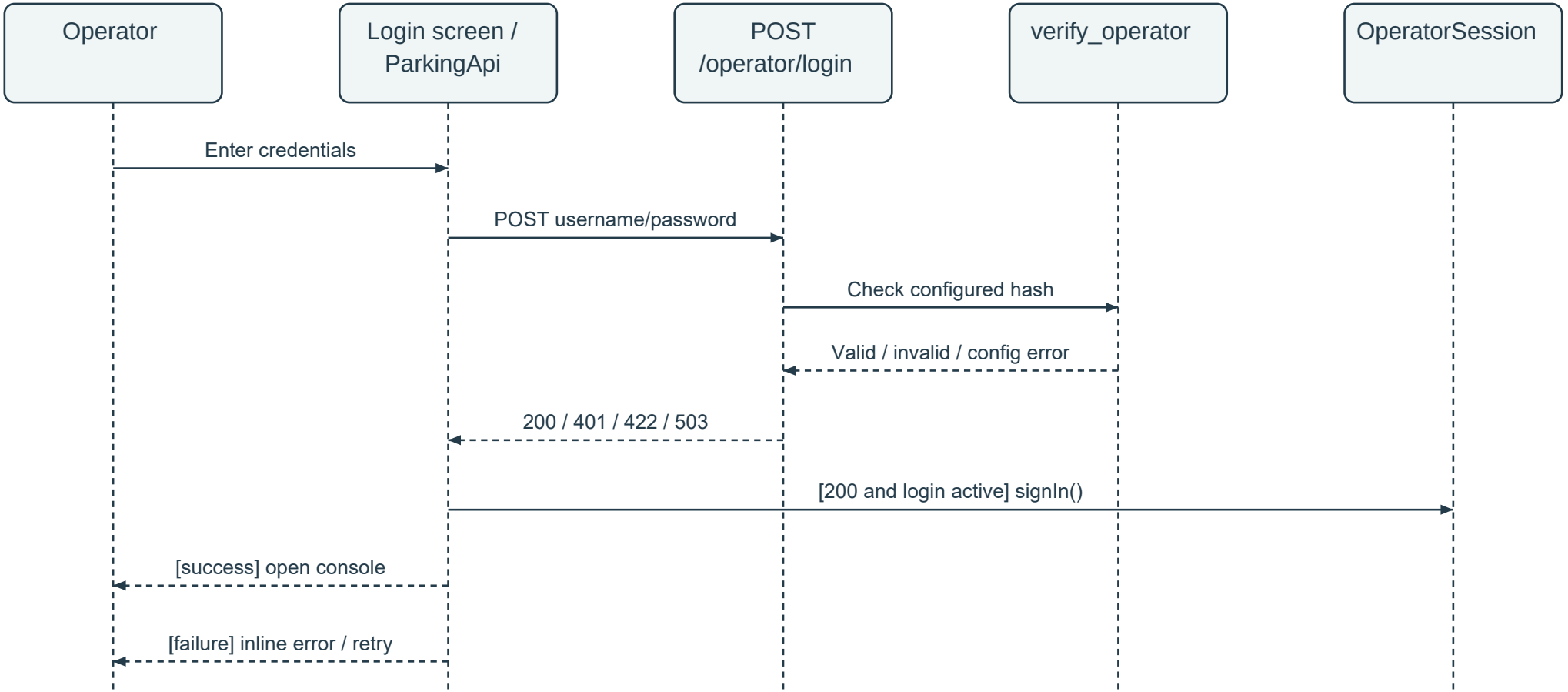



Shortlist uses Haversine distance first. DB transactions close before network calls. No automatic replacements.

The shortlist is chosen before route/model eligibility checks. A failed candidate is not replaced by an additional search. Probability requests can still be evaluated for a candidate whose quote fails; the final gate prevents it from ranking.

Implementation reference: backend/recommendation.py; comparison.py; model_adapter.py; Flutter ParkingApi

# Figure 12. UC-05 operator credential sequence

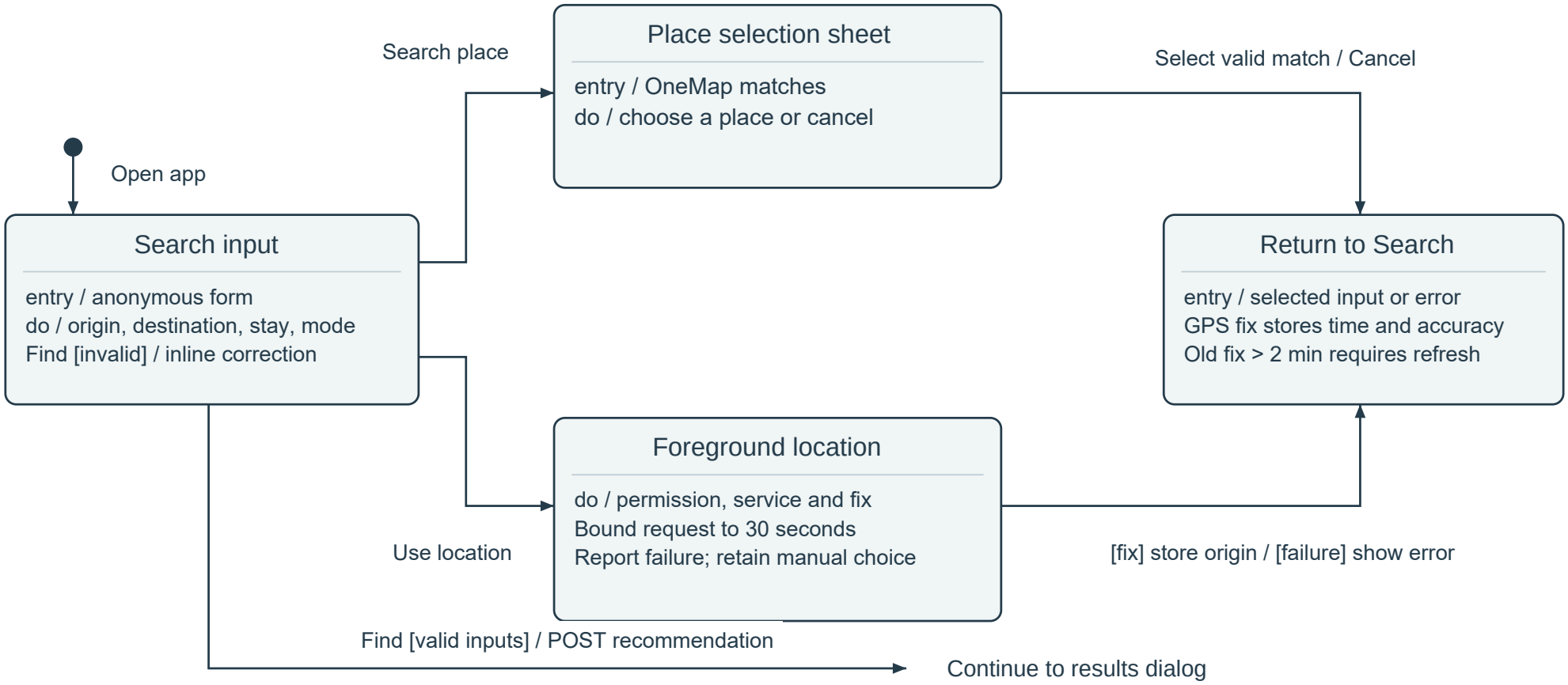



Empty fields fail locally. Repeated taps send one request. Leaving the screen discards late success.

Invalid schema input may return HTTP 422. Duplicate submissions are blocked and leaving the screen prevents a late response from establishing a new app session.

Implementation reference: operator_login_screen.dart; parking_api.dart; operator_session.dart; operator_auth.py

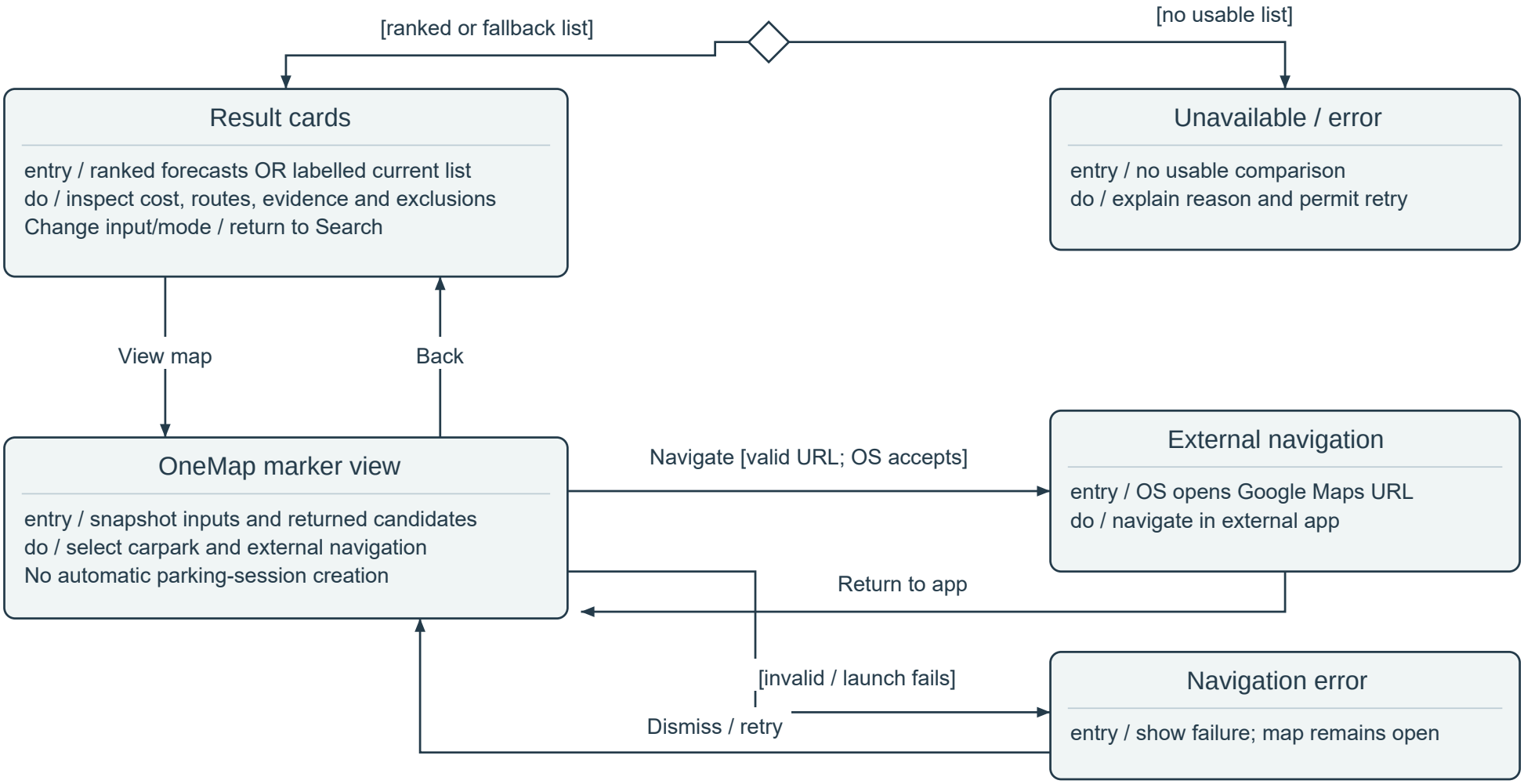
# Figure 13. Driver dialog map: input and search lifecycle




A foreground device fix is one origin input, not continuous tracking. A fix older than two minutes must be refreshed. Manual place selection remains available after location failure.

Implementation reference: carpark_search_screen.dart; device_location.dart

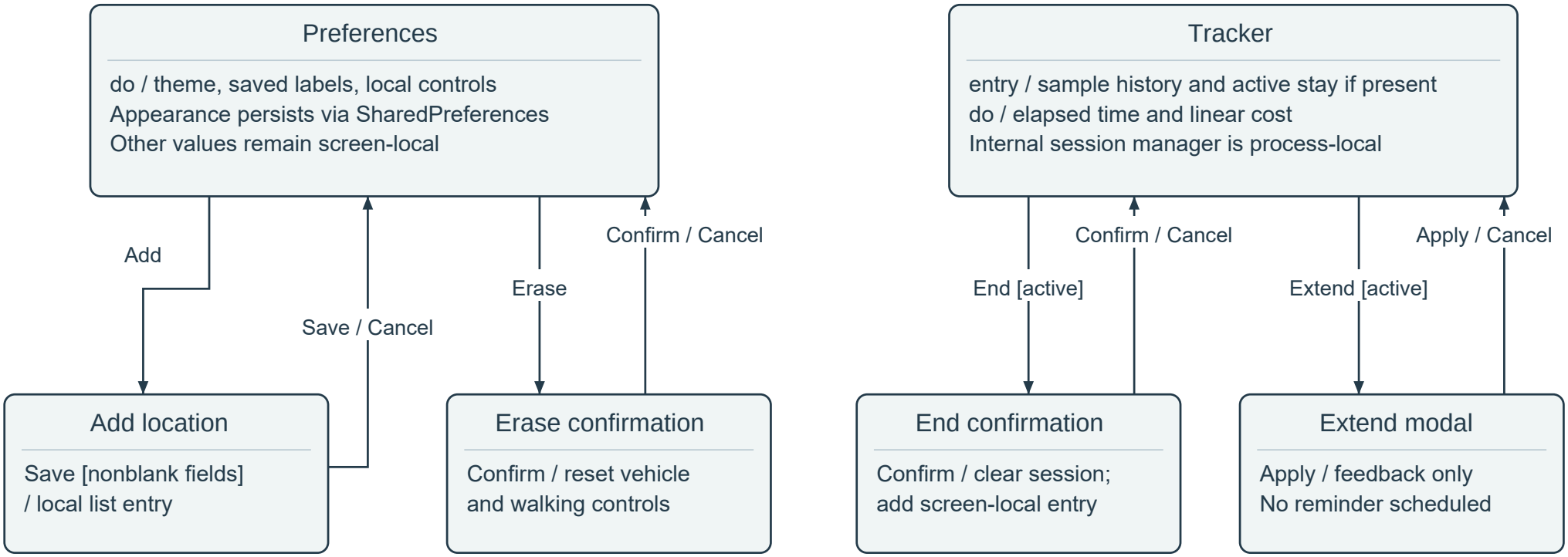
# Figure 14. Driver dialog map: results, fallback and navigation




Result cards are the current inspection boundary. There is no separate complete detail screen, report form or automatic start-parking transition. The map can display returned candidates, including candidates excluded from ranking.

Implementation reference: carpark_search_screen.dart; carpark_map_screen.dart; navigation_handoff.dart

# Figure 15. Driver dialog map: preferences and tracker

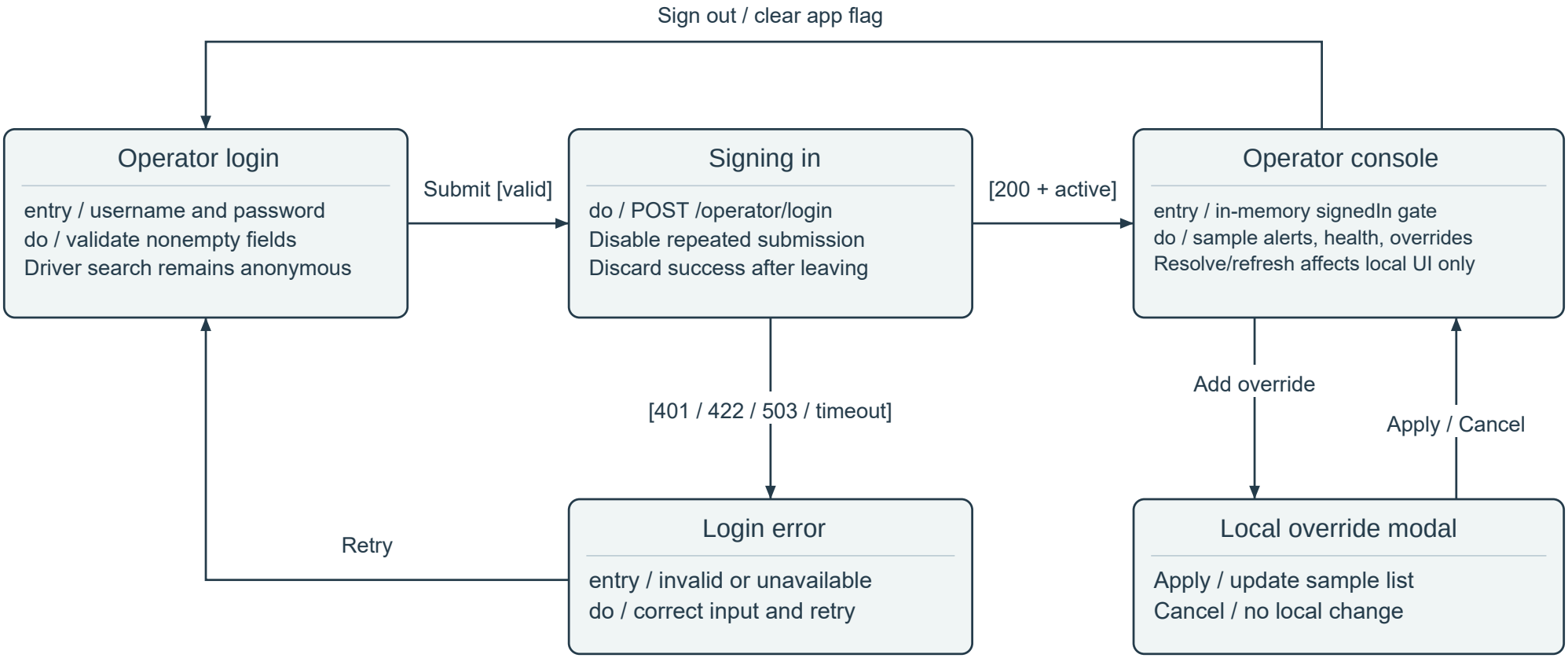



Bottom navigation opens each screen; leaving/recreating it can discard local sample edits.

This map includes prototype flows because they remain visible in the app. Their labels state the actual effect of each action; modal feedback is not modelled as durable storage or a completed notification operation.

Implementation reference: preferences_screen.dart; parking_session_tracker_screen.dart; local state services

# Figure 16. Operator dialog map: access and prototype console

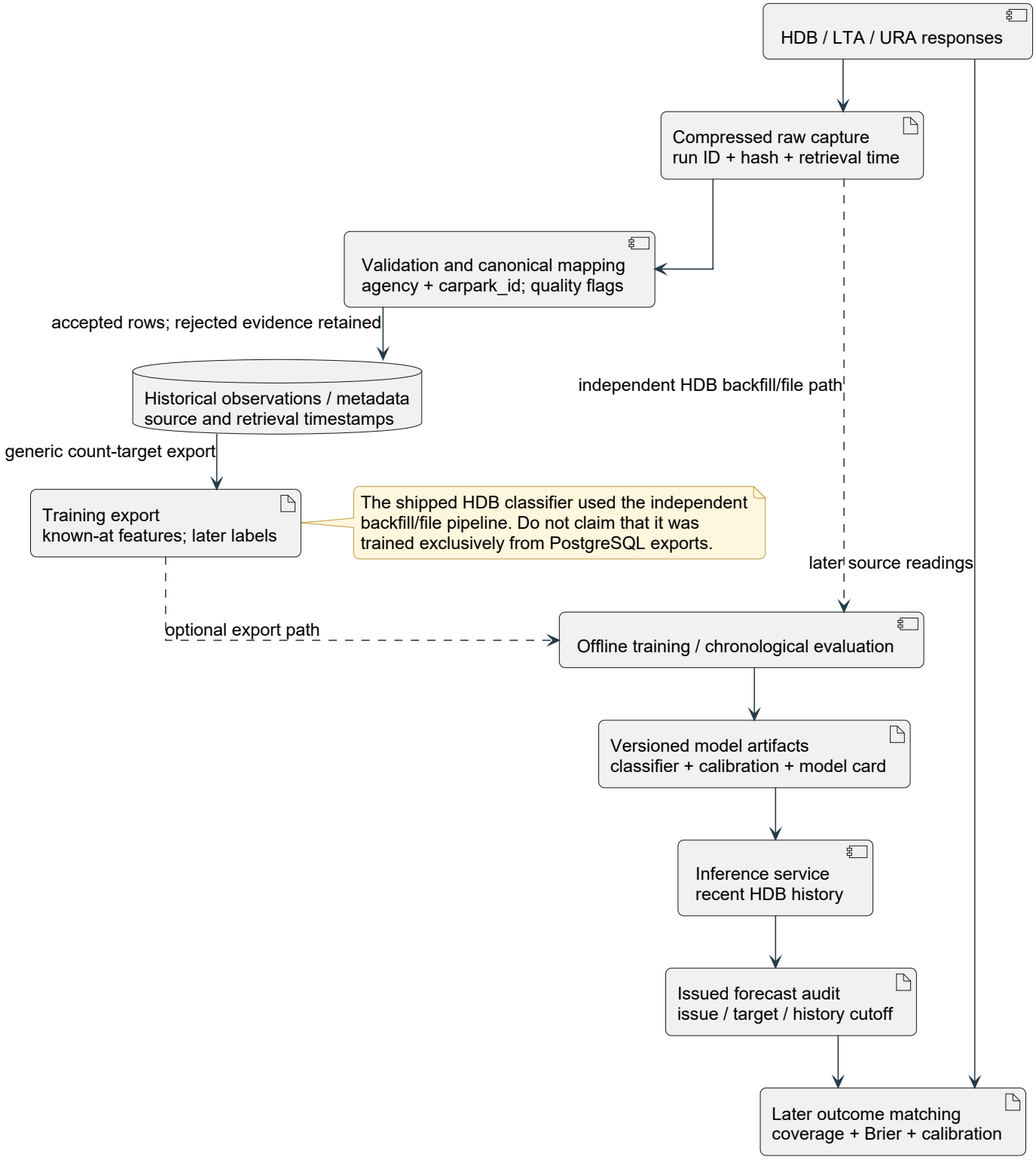



Admin tab opens console if already signed in; otherwise login. No server write authorization is issued.

The app gates console entry with a successfully verified credential. Its sample actions remain isolated from the recommendation database and do not constitute administrative authorization.

Implementation reference: operator_login_screen.dart; admin_panel_screen.dart; operator_session.dart

# Figure 17. Collection, cleaning and training evidence lifecycle




Recommendation inference does not retrain the classifier. Forecasts are recorded before target time, and later outcomes are matched with a declared tolerance. Provider feed evidence is not independently verified physical occupancy.

Implementation reference: data-pipeline collectors, metadata cleaner, export_training.py; ml training/service

# Recommendation scoring and eligibility

Eligibility is evaluated before scoring. Valid driving/walking routes, supported current tariffs, coherent identity/capacity evidence and an acceptable forecast are required. Enabled HDB forecasts use short mode at 15-90 minutes, recent source history no older than 15 minutes, consistent confidence/model metadata, and p_free >= 0.90.

| Mode | Cost | Walk | Availability | Drive |
|---|---|---|---|---|
| Best overall | 40% | 30% | 20% | 10% |
| Prioritise cheaper | 65% | 15% | 15% | 5% |
| Prioritise closer | 15% | 65% | 15% | 5% |

Each nonnegative factor becomes a bounded penalty. Let C be cost in cents, D the stay duration in minutes, W the walking route distance in metres, T the driving duration in seconds, and p the arrival free-lot probability.

| Penalty | Implemented formula |
|---|---|
| Cost | C / (C + 400 * D / 60) |
| Walking | W / (W + 500) |
| Driving | T / (T + 900) |
| Forecast availability | 1 - p |
| Current-only availability | 1 / (1 + current available lots) |
| Final score | 100 * (1 - sum(weight * penalty)) |

Higher scores rank first. Ties use walking then cost in Closer mode, cost then walking in other modes, followed by driving duration and canonical identity. Fixed scales avoid changing pairwise penalties when another candidate appears. Weights/scales are an explicit project policy, not claimed optimal driver preferences.

Current-only fallback is a separate list with a visible notice. It requires fresh source-timestamped positive current counts, usable routes/prices and an allowed missing-forecast reason. A known low probability is not promoted into fallback. A failed candidate is not automatically replaced.

# Evidence and data dictionary

| Field / concept | Meaning and constraint |
|---|---|
| Canonical identity | (agency, carpark_id). Source codes map only using exact reviewed identity or an explicit verified mapping. |
| latitude / longitude | Finite WGS84 decimal degrees. A single supported point is required for routing. Multiple/ambiguous geometries remain flagged. |
| lots_available / total_lots | Current counts. HDB capacity is synchronized with its reading; LTA/URA feeds do not provide equivalent total capacity or per-carpark source times. |
| evidence_at / retrieved_at | Observation/source time and retrieval time have different semantics. A retrieval proxy is labelled as such. |
| issued_at / arrival_at | Forecast issue time and candidate-specific target arrival. Preserve the service issue time; do not relabel it as request time. |
| history_cutoff | Latest source history used by the forecast; it must precede issue time and pass the freshness limit. |
| p_free / confidence | Probability of at least one free lot. High >= 0.99; Medium >= 0.90 and < 0.99; Low < 0.90. It is not a predicted count or reservation guarantee. |
| amount_cents / range | Full-stay supported estimate, source URL and schedule ID. Compare neighbouring-minute arrivals and score the higher supported quote. |
| schedule validity / review | Version valid_from/valid_until and review metadata. HDB/commercial review limit: 30 days; URA: 2 days. Changed facility flags and unresolved windows can block a quote. |
| ranking_exclusions | Machine-readable reasons a candidate did not qualify; unavailable measurements are not replaced with zeros. |
| Device-local state | Only appearance is durably stored in the current preferences implementation. Sessions, local saved labels and console samples do not form backend driver-profile tables. |

# Implementation and test traceability

| Requirement | Source components | Test evidence |
|---|---|---|
| UC-01, UC-02 | recommendation.py; comparison.py; pricing.py; scoring.py; CarparkSearchScreen; ParkingApi | backend/validation/test_workflow.py, test_lab4_controller.py, test_ranking_modes.py, test_pricing.py, test_current_fallback.py; Flutter parking_search_test.dart, parking_api_response_test.dart |
| UC-03, UC-19 | CarparkMapScreen; navigation_handoff.dart; device_location.dart | Flutter location_and_navigation_test.dart, map_stability_test.dart, device_location_test.dart |
| UC-05 | OperatorLoginScreen; OperatorSession; app.operator_login; operator_auth.py | test_operator_auth.py; Flutter operator_login_test.dart |
| UC-10 | collect_live.py; collect_periodic.py; puller.py; provider_puller.py; metadata_cleaner.py; export_training.py | test_collect_periodic.py, test_puller.py, test_provider_puller.py, test_metadata_cleaner.py, test_export_training.py, test_refresh_pricing.py |
| UC-11, UC-12, UC-14-18 partial UI | AppThemeController; PreferencesScreen; ParkingSessionManager; Tracker | Flutter widget_test.dart, frontend_lifecycle_test.dart, frontend_stress_test.dart. UI checks do not establish unimplemented durable storage/notifications. |
| UC-04, UC-13 and server admin istration | Deferred or sample-only components | No claim of complete backend acceptance coverage. |

The submission branch includes executable tests and dated result evidence. Test fixtures verify behavior and failure handling; live source availability, route/provider health, physical-device acceptance and prediction accuracy are separate checks. Model evaluation is retained under ml/reports/; the model is not retrained by building this document.

The final SRS should retain the partial/deferred requirement statuses in this document. Do not describe sample console health, feedback-only refresh, incomplete erasure or local session history as working persisted backend features.